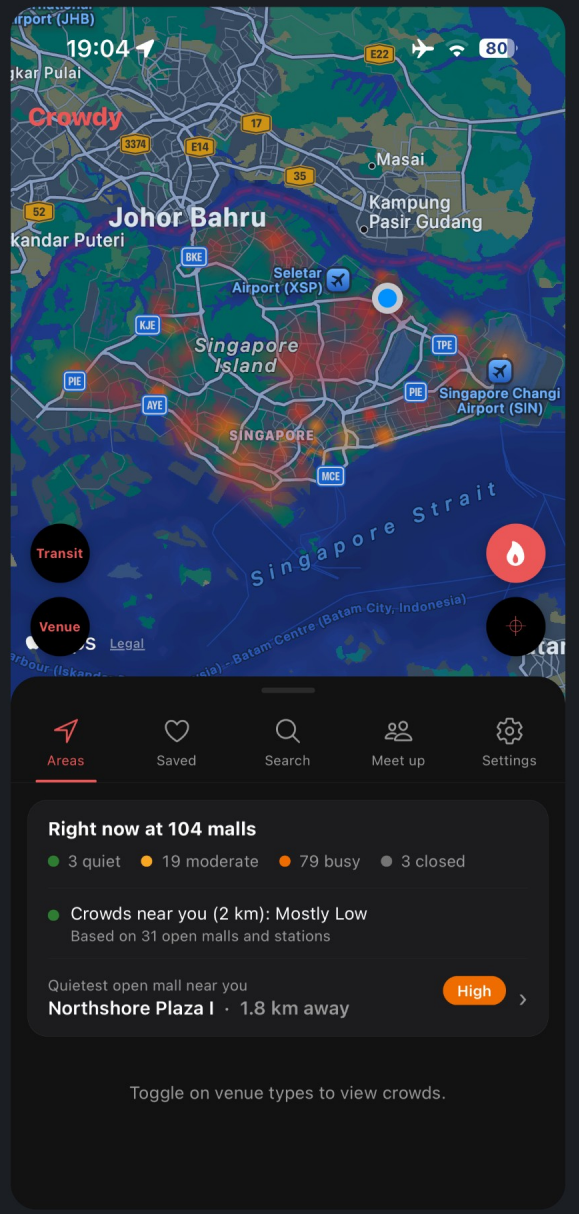
SC2006 SOFTWARE ENGINEERING · DEMO

Crowdy

Check how crowded Singapore's malls and transit stations are,
and find places for friends to meet.

SCSJ Group 3
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THE PROBLEM

You only find out a place is packed when you get there



Crowdy shows how busy 104 malls and every MRT/LRT station are, suggests quieter places nearby, and helps friends pick a fair place to meet.

LTA DataMall
Live crowd level at every station, every 5 minutes

Google Popular Times
Each mall's typical week, hour by hour, and its opening hours

OneMap
Routes and travel times for directions and meetups

Land Transport DATAMALL

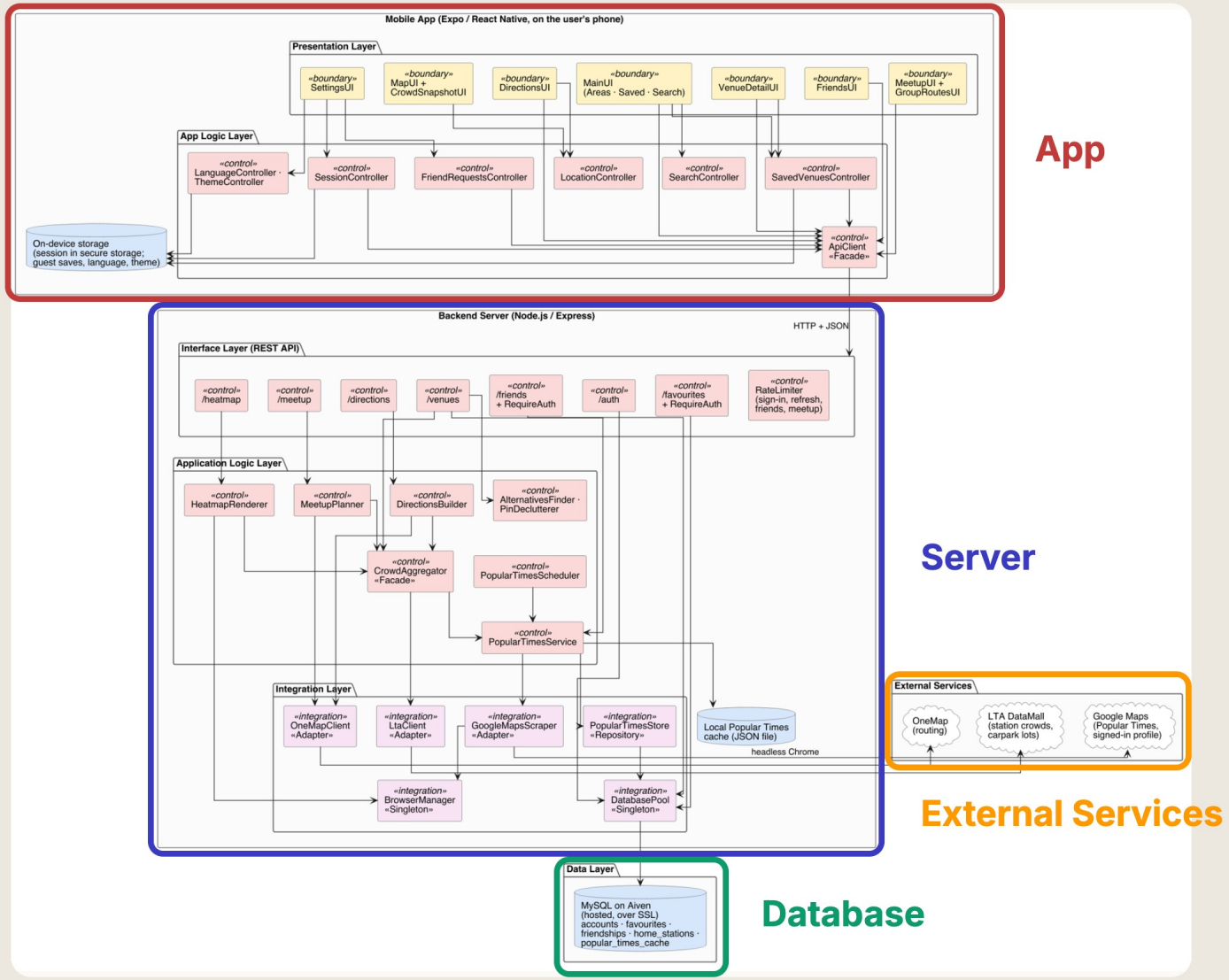
Google

OneMap Singapore

Individually, none of these sources answer: "Where should we go that isn't crowded?"
Crowdy merges them to ease an everyday annoyance in the small bustling city of Singapore.

DESIGN

Layered architecture



FEATURE · LIVE CROWD MAP

Every venue, coloured by how busy it is

The Right now card counts quiet, moderate and busy places and names the quietest mall near you.

Facade
CrowdAggregator merges all three sources into one venue list

Abstraction
Malls and stations share one Venue type, so every screen treats them alike

Fail-safe design
If a source fails, its last good data is kept, so the map never goes blank

FEATURE · VENUE PAGE

Typical crowds and quieter alternatives

The crowd for this hour, labelled as typical, the usual week as a chart, opening hours, and up to three quieter malls within 3 km.

Single responsibility
AlternativesFinder has one job: choosing the alternatives

Black-box testing
13 cases from equivalence classes and boundary values

Shared data cache
One shared copy of the collected data, with a local backup

FEATURE · GETTING THERE

Directions and search

Routes in four modes with the live crowd at each station you board, and search that forgives a one-letter typo.

Adapter
OneMapClient turns OneMap's format into our own route types

Reuse
Route crowd levels reuse the map's live station data

Basis path testing (white-box tested)
Search matching (tokenHits) has complexity 6, so 6 test paths

FEATURE · ACCOUNTS AND FRIENDS

Saved places, friends and security

Save places as a guest or to an account. Crowdy asks before adding a guest’s saves, and alerts you to friend requests.

Security
bcrypt password hashes, signed sessions, rate limiting per client

No hints to attackers
A wrong username and a wrong password get the same message

Observer
When friend requests change, every badge updates by itself

FEATURE · MEETUP PLANNER

A fair mall for the whole group

For 2 to 6 people, now or at a chosen time in the next week, ranked by fairness, total travel time or crowd.

Clear ranking rules
Least crowded only picks malls that stay close for everyone

Performance by design
Shortlist first, at most 8 route lookups at once, trips cached 6 hours

Tested
11 automated tests cover each priority and chosen times

ENGINEERING · QUALITY

Testing, traceability and workflow

50
server tests, all passing

24
app tests, all passing

0
critical security findings

Traceability
Requirement, use case, design class, code and tests all link up

Git workflow
Work on dev, reviewed feature branches, main is what we submit

Thank you

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